

1) Given the following unit sample response

- $h[n] = (n + 1)\alpha^n u[n]$

where $|\alpha| < 1$. Find the step response of the system.

2) Check whether the following signal is a power

signal($0 < P_\infty < \infty$) , energy signal($0 < E_\infty < \infty$) or

both.

- $y(t) = e^{-|\alpha t|} \cos(\beta t)$

3) Calculate the convolution of 2 following signal

- $x(t) = u(t) - 2u(t - 2) + u(t - 5)$

- $h(t) = e^{2t} u(1 - t)$

4) Consider a transformation, denoted by T , that operates on input signals. This transformation is defined by the following two properties:

1. Basic Transformation Rule: For any input of the form $x(t) = t^k$ (where k is a non-negative integer), the corresponding output is $y(t) = \cos(kt)$. That is:

$$T\{t^k\} = \cos(kt)$$

2. Additivity for Power Function Combinations: The transformation satisfies the additivity and homogeneity properties for inputs that are algebraic sums of power functions (t^k). Specifically, if the input is $x(t) = c_1 t^{k_1} + c_2 t^{k_2} + \dots + c_n t^{k_n}$, then the output is given by:

$$T\{x(t)\} = c_1 T\{t^{k_1}\} + c_2 T\{t^{k_2}\} + \dots + c_n T\{t^{k_n}\}$$

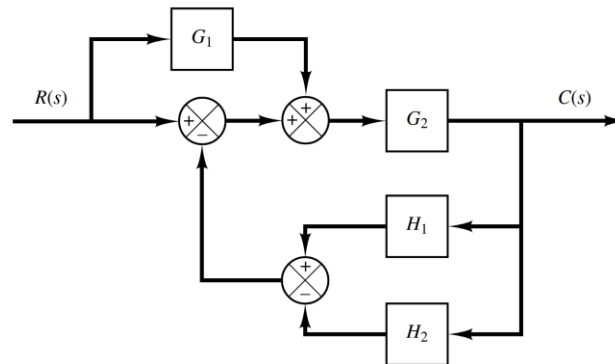
Note: This transformation is not necessarily a fully linear system, but it specifically exhibits additivity and homogeneity for this class of inputs.

Given the definition above, find the response of the transformation T to the following input:

$$x(t) = \frac{1 + t^{10}}{1 + t^2}$$

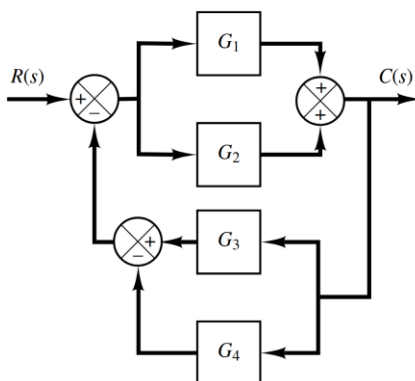
5) Transfer function for a close loop block diagram is basically the ratio of the output signal to the input signal denoted in the following question by $\frac{C(s)}{R(s)}$.

Find the closed loop transfer function for the following block diagram



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7) Check the linearity, causality, time variation ,and stability of the following system. Give a full solution.

- $y(t) = x(2) + x(t - 6)$

8) Check whether the following signal is stable or not. Give a full solution.

- $h(t) = te^{-t}u(t)$

9) Is the following statement correct or not? Give a full solution.

- $x[n] * (h[n]g[n]) = (x[n] * h[n])g[n]$